Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, turning owners' growth goals into site analyses and identifying a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price) and a margin model that turned client profit goals into break-even revenue and volume targets.
- Estimated the incremental revenue of new sites with Double ML and causal forests on a 12-year county panel, stating identification assumptions and checking robustness with difference-in-differences.
- Back-tested demand predictions against real sites (bias, error, rank correlation, 90% interval coverage) and built low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR, CVaR).
- Built reproducible Python pipelines with data-quality checks and 150+ unit tests, cutting turnaround by 91%, and documented each method for reuse.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Owned the consumer funnel KPI framework linking gross lead rate (GLR) to member rate (activation and active membership) and cost per member (CAC), with strategy, finance, and engineering leadership.
- Redefined GLR for a new client missing key lead markers, working with data engineering to date each lead by last response, credit it to its campaign, and flag broker influence.
- Built a causal graph (DAG) from historical data showing GLR drives member rate, which drives cost per member, focusing modeling and causal tests on GLR.
- Built an in-house experimentation tool that data science and strategy used to validate lift by segment and set campaign budgets, with CUPED cutting metric variance by 20%.
- Built uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, cutting ad spend by $10K per campaign.
- Found one client's "Opportunist" segment drove GLR and reallocated 12% of campaign budget toward it.
- Linked the monthly retention survey and member funnel to retention, finding claims and ticket complaints drove churn.
- Owned monthly monitoring of up to 10 production models for leadership and a company-wide self-service dashboard on curated Snowflake and Databricks tables.
- Re-architected the churn prediction platform on Spark and Databricks, cutting runtime by 48% and memory by 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation, and drift monitoring for search and pricing models, and mentored the team.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) built on item-level cost tables, securing a stable 18% contribution margin.
- Cut search dissatisfaction by 12% and raised platform engagement by 16% with custom Elasticsearch indexing and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Defined and reported the core campaign metrics (GLR, member rate, cost per member).
- Reverse-engineered a funnel model to turn financial mandates into team targets: a $1.00 indexed cost per member required GLR of at least 14.2%, with each +1 GLR point lowering the index by about 0.07.
- Optimized GLR to drive a 24% increase in paid member conversions within an average 60-day horizon.
- Built an out-of-sample leading indicator test showing GLR predicts member rate 60 days ahead ($R^2 = 0.28$).
- Allocated campaign spend across prospect audiences by eligibility and historical response rates.
- Built XGBoost response models explained with SHAP, lifting response rates by 3%, and automated Power BI reporting, saving 60 minutes daily.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization and operator theory, and taught statistics to classes of 75-200 students.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show when LTV = ARPU / churn breaks down, estimating LTV from survival curves instead.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Tested 1- and 7-day player retention with sample sizing, significance testing, and practical significance.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse on 790K claims, a fraud detector with 81% precision, and forecasts 41% more accurate than a seasonal benchmark.

Starbucks Rewards: Predicting Offer Responses github.com/beauvilerobed/predict-consumer-responses

- Predicted how rewards members respond to offers with XGBoost trained and deployed on AWS SageMaker.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Trees, Metric Definitions, Consumer Funnel (Acquisition, Activation, Engagement, Retention), CAC, LTV / ARPU, Sensitivity & Scenario Analysis, Target Setting, Budget Allocation, Forecast Backtesting

Causal & Experimentation: A/B Testing, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs)

Modeling: Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, SHAP, Model Monitoring

Tools: SQL, Python (pandas, statsmodels, scikit-learn, econml), Snowflake, Databricks, Spark, dbt, AWS, Azure, Power BI, Tableau, Git